

Doubled Pedagogical Agents in an Immersive Virtual Environment

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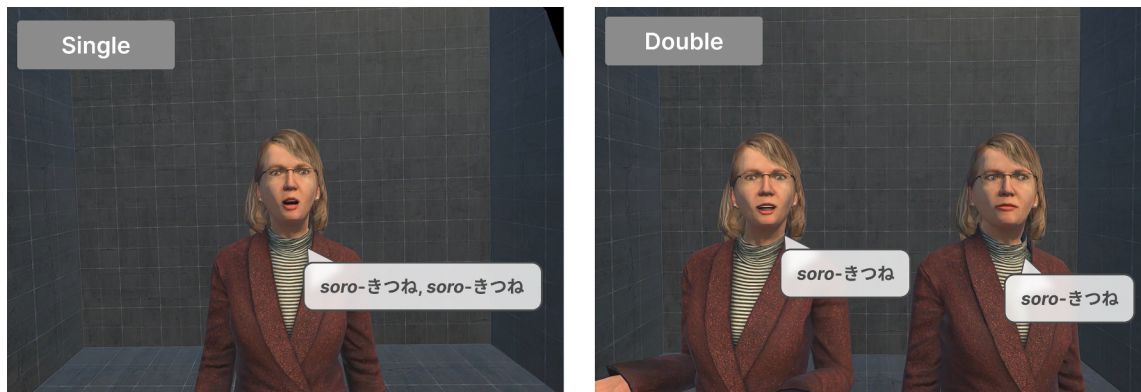
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Figure 1: Experimental conditions and system overview. (Left) In the Single condition, a single pedagogical agent repeated each Tagalog-Japanese word pair twice. (Right) In the Double condition, two agents each spoke the pair once to maintain an equivalent total number of exposures.

ABSTRACT

To enhance the effectiveness of remote lectures and asynchronous education, researchers have investigated how the design of virtual pedagogical agents influences learning. While prior research has primarily focused on the effects of agent appearance, this study examined the impact of the number of agents within an immersive virtual environment on memory and agent perception. We hypothesized that presenting information from two pedagogical agents would improve performance on memory tasks compared with a single agent, but would also elicit uncanniness due to their identical appearance, known as the clone devaluation effect. We conducted a preliminary study, and results from eight participants suggested that the order of conditions might reverse effects on recall accuracy in the within-participants design. These findings indicate that the experimental design should be carefully reconsidered, for example, by including a practice trial to familiarize participants with the memory task and virtual environment.

Index Terms: Pedagogical agent, virtual human, learning, memory, virtual reality.

1 INTRODUCTION

Approaches that display virtual agents as on-screen instructors have been proposed to enhance the effectiveness of remote lectures and online learning, a concept known as pedagogical agents [6]. It is well known that these virtual agents are more than on-screen images; they establish social relationships with users and significantly influence their perceptions and cognition. For instance, Baylor and Kim [1] investigated the effects of agent attributes, such as ethnicity, gender, visual realism (realistic vs. cartoon-like), and role

(expert vs. peer), on learning. Their findings suggested that greater agent realism correlates with improved learning outcomes. Similarly, Mizuho et al. [10] conducted a memory task in which characters with human, animal, robot, and extraterrestrial appearances spoke target words. They suggested that human agents might improve memory performance more effectively than other representations. Thus, researchers have attempted to support learning and memory by manipulating agent representations to alter the social relationship established with the user.

Traditionally, research on agents for educational and learning purposes has primarily focused on agents displayed on two-dimensional (2D) monitors. However, since the late 2010s, with the widespread adoption of affordable head-mounted displays (HMDs), the use of pedagogical agents in immersive virtual environments using virtual reality (VR) has begun to attract attention [7, 14]. It is well established that increased immersion enhances the social presence of agents [3]; thus, it is anticipated that pedagogical agents in immersive virtual environments will have a stronger impact on learning and memory than on-screen agents. As this field is still in its nascent stage, prior studies remain limited and have mainly investigated the effects of agent appearance. For instance, studies have examined the effects of manipulations such as visual realism [14], appearance congruent with the user's gender (e.g., an agent modeled as a young female researcher for female students) [7], and the diversification of appearance in repetitive learning (e.g., doctors, soldiers, and businesspersons) [11].

Thus, research on pedagogical agents in VR has focused on the effects of appearance, largely as an extension of 2D agent research. However, unlike 2D environments, VR enables manipulation of factors beyond appearance, such as spatial arrangement and bodily movement. Therefore, we considered that exploring methods and effects unique to VR would yield insights beyond a mere extension of 2D agent research. In this study, we focus on the number of agents, a factor known to enhance agent effectiveness in social robotics research. Prior studies suggest that conditions using two robots improve learning motivation [17] and the retention of motor

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skills [16] compared to a single robot. Based on these findings, we hypothesized that using two pedagogical agents in VR would result in higher performance on memory tasks than using a single agent. To test this hypothesis, we developed a VR system in which pedagogical agents present Tagalog-Japanese word pairs as memory targets and conducted a preliminary study with eight participants. The objectives of this preliminary study were to adjust the difficulty of the memory task (e.g., to avoid floor and ceiling effects), verify the overall system operation, and estimate the effect size. We also conducted interviews with participants to gather qualitative insights regarding the design of the two pedagogical agents.

2 EXPERIMENT

2.1 Experimental Design and Hypotheses

Participants performed a memory task in a VR environment, where they memorized Tagalog-Japanese word pairs presented vocally by virtual agents. The experiment followed a within-participants design with two conditions: a Single condition with one virtual agent and a Double condition with two agents (Figure 1). The evaluation metrics included the recall accuracy of an immediate test and questionnaire-based measures of the agents' social presence and uncanniness. To control factors other than the number of agents, the agents' appearance and voice were identical across both conditions. Furthermore, the amount of information provided by the agents, such as instructions and words, was equivalent between conditions. Specifically, in the Double condition, the two agents took turns speaking the same dialogue script used in the Single condition. Regarding physical movements, the two agents in the Double condition alternately performed the same animations used in the Single condition. The presentation order of the conditions was counterbalanced across participants.

Previous research suggests that the number of social agents can amplify their social influence on users [16, 17]. Based on these findings, we hypothesized the following:

H1 The recall accuracy in the memory task will be higher in the Double condition than in the Single condition.

Additionally, the "Clone Devaluation Effect" [19] suggests that encountering multiple entities with an identical appearance can elicit a sense of eeriness. Consequently, we proposed a second hypothesis:

H2 Perceived eeriness will be higher in the Double condition than in the Single condition.

By synthesizing memory performance data and perceptual evaluations, we aim to clarify how the number of agents influences memory performance.

2.2 Participants

Eight healthy individuals (seven males and one female, mean age $22.6 \pm 1.30(SD)$) participated in the study. To minimize the influence of prior familiarity with the target Japanese nouns on memory performance, all participants were recruited as native Japanese speakers. Each participant was compensated with a 1500-yen Amazon gift card. The experimental protocol was approved by the local ethics committee of the Graduate School of Information Science and Technology, the University of Tokyo (UT-IST-RE-250604).

2.3 To-be-remembered Materials

Participants memorized two sets (List A and B) of 10 Tagalog-Japanese word pairs. Following previous studies [11, 12], Tagalog was selected because most participants have no prior knowledge of it, minimizing confounding variables. Given that the previous study [11] reported an accuracy of about 30% (6 words) for 20-word lists in immediate tests, we determined that 10 words would

offer suitable difficulty, yielding an average accuracy of approximately 60%. To control for word length, which could affect the ease of memorization, 20 Tagalog nouns with two syllables each were selected from a dictionary. Examples include *ahas* (snake) and *bilog* (circle). The assignment of word sets to each condition was counterbalanced across participants.

2.4 Experimental System

The VR environment was rendered using Unity 2021.3.26f1. Participants were immersed in the environment wearing an HMD (Meta Quest 3) and headphones (Sony WH1000XM3). To facilitate natural attention toward the virtual agents, we used a simple virtual room without furniture. The participants' bodies were represented as self-avatars within the VR environment. We selected neutral avatars from the open-source Microsoft Rocketbox Avatar Library [2], specifically *Male_Adult_08* for male participants and *Female_Adult_01* for female participants. The full-body movement of the self-avatars was implemented via inverse kinematics (IK) using the RootMotion FinalIK plugin¹, based on the position and rotation of the HMD and hands.

The pedagogical agents were also selected from the Microsoft Rocketbox Avatar Library. We adopted *Business_Female_02*, which received positive feedback in a previous study [11] as being perceived as a plausible instructor. The agents' physical movements were implemented using suitable animations (e.g., idling) from the same library. The voices for both Tagalog and Japanese were generated using Ondoku3 software², and the agents' mouths moved in synchronization with the audio via the OVRlipsync plugin³. Virtual speakers were placed at the mouth position of each agent with spatial audio settings. This implementation ensured that participants could naturally perceive which agent was speaking, especially in the Double condition.

2.5 Procedure

Participants first provided their consent via an online form. After receiving detailed instructions about the procedure, they completed the Simulator Sickness Questionnaire (SSQ) [5] to assess baseline VR sickness. They then put on the HMD and headphones to immerse themselves in the VR environment. The pedagogical agent(s) gave a brief greeting (e.g., "Hello, [Participant Name]"), followed by a 30-second period where participants were instructed to look around the room to familiarize themselves with the environment and the agents.

Next, the agent(s) presented 10 Tagalog-Japanese word pairs, and participants memorized as many Japanese translations as possible. Each pair was presented twice at 5-second intervals, with the 10 pairs shown in a continuous, randomized order. In the Single condition, one agent repeated each pair twice, while in the Double condition, the two agents each spoke the pair once (Figure 1).

After removing the HMD and headphones, participants performed a 30-second distraction task consisting of simple arithmetic (e.g., $4 + 2 \div 2$) on paper to prevent mental rehearsal, as in the earlier studies [4, 9]. Subsequently, they took a recall test using a monitor. Tagalog words were played through a speaker (Bose Soundlink II), and participants responded with the Japanese translation orally within a 5-second time limit per item. The 10 items were tested in a randomized order, with a countdown timer displayed on the monitor. Finally, participants completed a questionnaire measuring the agents' social presence and uncanniness [20].

The cycle of memorization, interval, recall test, and questionnaire was repeated for the other condition. The trials under the first

¹<https://assetstore.unity.com/packages/tools/animation/final-ik-14290>

²<https://ondoku3.com/>

³<https://developer.oculus.com/downloads/package/oculus-lipsync-unity/>

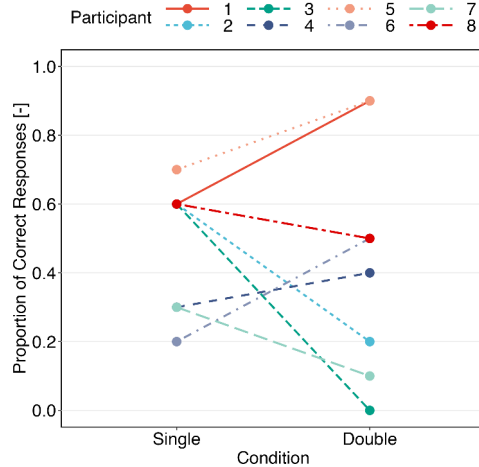


Figure 2: Recall accuracy of individual participants. Half of the participants performed better in the Single condition, while the other half showed higher accuracy in the Double condition.

and second conditions are referred to as Trial 1 and Trial 2, respectively. Finally, participants completed the SSQ and the Igroup Presence Questionnaire (IPQ) [15] to evaluate their subjective sense of presence in VR. This was followed by semi-structured interviews regarding the memory task (e.g., “Were there any words you already knew?”) and the virtual agents (e.g., “Did you notice any differences between the single and double agent conditions?”). After the interviews, participants received their compensation and were dismissed. The experiment lasted approximately one hour.

3 RESULTS AND DISCUSSION

Due to the preliminary nature of this study, we report only the primary results: recall accuracy in the memory task and subjective ratings of eeriness. Additionally, we did not perform statistical hypothesis testing in this analysis; instead, we reported effect sizes where necessary to avoid misinterpretation due to insufficient detection power, while gaining insights for future study.

3.1 Recall

Except for one participant, none of the participants reported having any prior knowledge of the target words. Since one participant reported knowing the word *kahon* (box), we calculated their recall accuracy based only on the remaining nine words and included this data in the analysis, following the procedure in a previous study [11].

Figure 2 shows the recall accuracy. Half of the participants (ID: 1, 4, 5, 6) performed better in the Double condition, while the other half (ID: 2, 3, 7, 8) performed better in the Single condition. Since the trends differed among participants, it is possible that factors other than the number of agents contributed to the recall accuracy. In the following, we focus on the possibility of order effects and conduct an exploratory analysis.

Figure 3 shows the recall accuracy for each trial (Trials 1 and 2). In Trial 1, the Single condition had higher recall accuracy compared to the Double condition (Cohen’s $d = 1.20$). On the other hand, in Trial 2, the Double condition showed higher recall accuracy compared to the Single condition (Cohen’s $d = 0.65$).

These results could potentially be interpreted based on the relationship between cognitive load and social facilitation/inhibition [8]. When cognitive load is low, the presence of agents enhances task performance (social facilitation). Conversely,

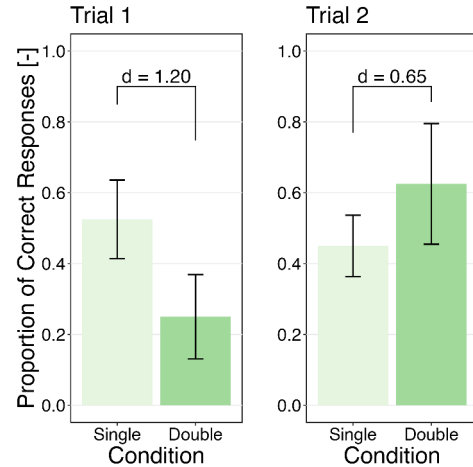


Figure 3: Recall accuracy for each trial. The Single condition showed higher accuracy in Trial 1, whereas the Double condition showed higher accuracy in Trial 2.

when cognitive load is high, the effect is reversed, and their presence hinders performance (social inhibition). Similarly, in this study, the presence of two agents may have negatively affected accuracy in Trial 1, where participants were unfamiliar with the task and cognitive load was presumably high. In Trial 2, however, as participants became accustomed to the task and cognitive load decreased, the two agents may have had a positive effect. Based on this discussion, future experiments could introduce a practice trial to mitigate order effects. Alternatively, testing with two levels of task difficulty (easy and difficult) might provide deeper insights into the relationship between cognitive load and the observed effects.

3.2 Perceived Eeriness

Figure 4 shows the subjective ratings of eeriness. A trend was observed where the Double condition had higher scores than the Single condition (Cohen’s $r = 0.29$). Since this eeriness could potentially affect memory performance, we will continue to carefully observe its influence in future experiments. Alternatively, to avoid uncanniness, we could consider using agents with different appearances or using agents that do not cause discomfort when having the same appearance, such as social robots [13] and animals [18].

3.3 Qualitative Feedback From Participants

Regarding the design of the doubled pedagogical agents, seven of the eight participants suggested using distinct appearances. Specific candidates included a combination of a foreigner and a Japanese person tailored to the nature of the memory task, a male-female pair as an analogy for parents, or agents wearing different-colored clothes. It was particularly interesting that some noted the familiarity of the teaching style in Japanese English education, which involves two instructors: a native speaker and a Japanese teacher. Regarding the agents sharing the same voice, many participants found the visual discomfort more prominent and were not bothered by the voice. However, it was pointed out that it would be strange if they had different appearances but the same voice. Additionally, some noted that having two instructors is uncommon in daily life, and suggested alternatives such as one instructor and one classmate. There were also comments that interaction between the two agents would be beneficial. How these designs contribute to memory performance remains unclear and is promising topics in the future.

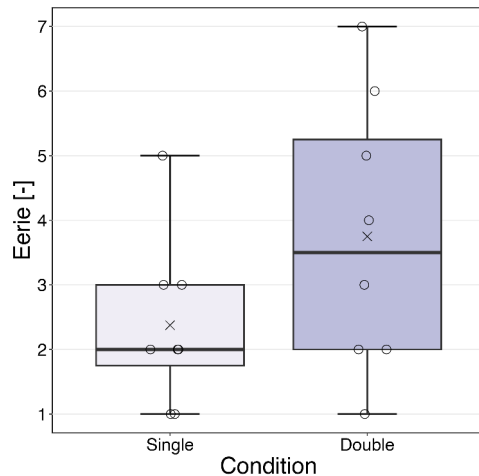


Figure 4: Perceived eeriness. Scores in the Double condition tended to be slightly higher than those in the Single condition.

4 CONCLUSION

To improve the effectiveness of learning support systems using virtual pedagogical agents in VR, this study focused on the effect of the number of agents and compared a Single condition with one agent and a Double condition with two agents in a within-participants design. A preliminary study with eight participants suggested that the expected effects might be influenced by the order of conditions, and it may be related to cognitive load. We will continue our investigation by introducing practice trials for the memory task, manipulating task difficulty, and considering a between-participants design instead of a within-subjects design.

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